

5 The fruit fly, *Drosophila melanogaster*, feeds on sugars found in damaged fruits.

A fruit fly with normal features is described as wild type. It has a grey body and its wings are longer than its abdomen. The genes for body colour and wing length are located on different chromosomes.

A fruit fly with mutations in these two genes has a black body and short wings.

Fig. 5.1 shows a wild type fruit fly and a mutant fruit fly.

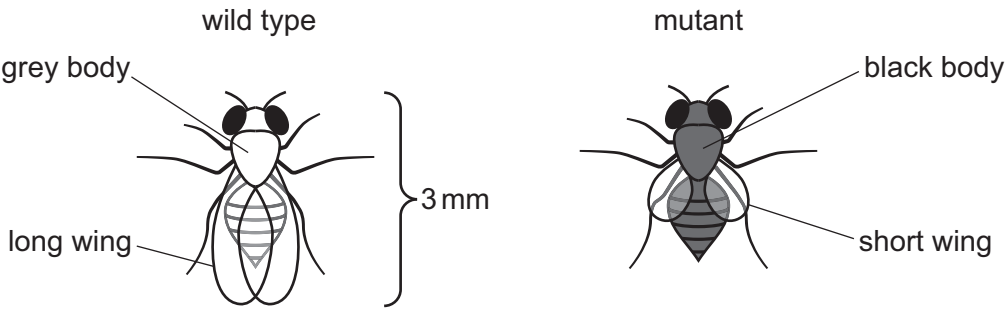


Fig. 5.1

- (a) Fruit flies were first used for genetic crosses by Thomas Morgan in 1908. They are one of the most studied animals in current genetic research.
- Male fruit flies are easily distinguished from female fruit flies.
 - Fruit flies have a short life cycle and a female can lay hundreds of eggs in a few days.
 - Some genes for development and cell signalling in fruit flies are similar to those of humans.

Suggest why fruit flies are still used in genetic crosses.

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- (b) When a wild type fruit fly was crossed with a mutant fruit fly with a black body and short wings, all the F₁ offspring had grey bodies and long wings.

Using appropriate symbols, complete Fig. 5.2 to show the expected results of a cross between two of these F₁ fruit flies.

symbols:

F₁ phenotypes: grey body × grey body
 long wing long wing

F₁ genotypes:

*Punnett
square*

offspring phenotypes:

phenotypic ratio

Fig. 5.2



- (c) Describe how you would determine the genotype of an F₂ fruit fly with a grey body and long wings.

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